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Histology

A Comprehensive Illustrated Study Guide

The microscopic anatomy of the four basic tissues

PART ONE · Foundations

Introduction to histology · microscopy · tissue processing & staining

PART TWO · The Four Basic Tissues

Epithelium · connective tissue · bone & cartilage
Muscle · nervous tissue

An original study guide by the EverythingABUAD team.

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HOW TO USE THIS GUIDE

Each section opens with a short orientation paragraph, then builds up detail in prose and tables. **Crimson bold** marks key terms worth memorising; shaded boxes flag high-yield facts and clinical correlates. Comparison tables are designed to be revised directly — cover one column and recall it from the other.

1 Introduction to Histology

Histology is the study of the tissues of the body — how cells and their surrounding material are arranged to build the organs, and how those arrangements serve the body's functions. It is, in effect, anatomy and physiology examined at the microscopic scale, and it spans every aspect of tissue biology from the individual cell up to the working organ.

1.1 What a tissue is made of

Every tissue is built from two interacting components that must be considered together. The first is the **cells** themselves. The second is the **extracellular matrix (ECM)** — a set of molecules the cells secrete into the spaces around them. Far from being inert packing, the matrix is highly organised and assembles into complex structures that give a tissue its mechanical properties and help route materials through it. The two components are in constant dialogue: the cells build and remodel the matrix, and the matrix in turn influences how the cells behave.

The ECM has two headline roles. It provides **mechanical support**, holding cells in place and giving the tissue strength and shape, and it acts as a medium for **transport**, allowing nutrients, wastes and signalling molecules to move between cells and the blood supply.

1.2 Why histology matters

Understanding normal tissue architecture is the reference point against which disease is recognised. Studying histology lets us understand the ways tissues are altered by disease, how cells behave and reproduce, and how whole organs function. Because so many diseases announce themselves first as microscopic changes, tissue analysis underpins a wide range of practical work.

- **Diagnosis** — examining biopsy and cytology specimens to identify disease, including the grading of tumours.
- **Autopsy** and **forensic investigation** — establishing cause of death and interpreting tissue evidence.
- **Research** — investigating how tissues develop, adapt and respond to treatment.

1.3 The four basic tissues

However complex an organ appears, its structure resolves into just four fundamental tissue types. Every later section of this guide examines one of them in turn, so it is worth fixing the overview first.

Basic tissue	Defining feature	Principal roles
Epithelial	Sheets of tightly joined cells covering surfaces and lining cavities; avascular	Protection, secretion, absorption, filtration, sensory reception
Connective	Few cells scattered in abundant extracellular matrix	Support, binding, transport, storage, defence, repair
Muscle	Elongated contractile cells rich in actin and myosin	Movement, posture, propulsion of contents, heartbeat
Nervous	Excitable neurons plus supporting neuroglia	Receiving, processing and transmitting electrical signals

KEEP IN MIND

Organs are mixtures. A single organ — the wall of the intestine, say — typically contains all four basic tissues layered together, which is why a histological slide usually shows connective tissue sandwiched between epithelium and muscle.

2 Microscopy

Tissues are far too small to study with the naked eye, so histology depends entirely on the microscope. Microscopy is the technical field of using a microscope to view objects, and detail within objects, that unaided vision cannot resolve. Different designs suit different jobs — some are built for routine biological and teaching work, others for metallurgy or for imaging three-dimensional surfaces — but all fall into two broad families.

2.1 The two great families

Optical (light) microscopy uses visible light and a system of glass lenses to magnify the image of a small sample. It is fast, adaptable and shows specimens in real colour. **Electron microscopy** instead works on the principle of a beam of electrons rather than light, and is the most advanced imaging approach in modern science; because electrons have a far shorter wavelength than light, it resolves structures down to the nanometre scale.

Feature	Light microscope	Electron microscope
Illumination	Visible light	Beam of accelerated electrons
Lenses	Glass lenses	Magnetic coils (electromagnetic lenses)
Resolution	Lower (limited by wavelength of light)	Very high — magnification into the nanometre range
Image	Direct, real colour	Captured as electron micrographs by specialised cameras
Typical use	Routine histology, teaching, live/coloured samples	Ultrastructure, metals and crystals, quality control

The metal filament used to generate the electron beam is typically tungsten. A high voltage excites its electrons into a continuous stream that serves as the 'illumination'. Magnetic coils then focus this beam onto the sample; as the current rises the magnetic lens strengthens. Because the beam cannot pass through a glass lens, the whole optical system must be magnetic.

2.2 Types of light microscope

Within the optical family, several variants tune the illumination or optics for particular tasks:

- **Simple microscope** — a single convex lens of short focal length. Placing the sample within the focus gives a virtual, erect, magnified image at the near point of the eye. Used by watchmakers, jewellers, dermatologists and in teaching labs.
- **Compound microscope** — combines two optical parts, an objective lens and an eyepiece (ocular). The objective forms a primary image inside the tube which the eyepiece magnifies again, so the two lenses multiply their power. Used to study bacteria and viruses, in forensic labs and in metallurgy.
- **Bright-field, phase-contrast, polarising, reflected-light, fluorescence and confocal** microscopes — specialised light designs for staining patterns, transparent living cells, crystalline materials, opaque surfaces, labelled molecules and optical sectioning respectively.

2.3 Types of electron microscope

- **Transmission EM (TEM)** — the beam passes through an ultrathin section to reveal internal ultrastructure.
- **Scanning EM (SEM)** — the beam scans a surface to build a three-dimensional image of its topography.
- Further variants include the reflection EM (REM), the scanning-transmission EM (STEM), the low-voltage EM (LVEM) and the scanning tunnelling microscope (STM).

2.4 Other specialised microscopes

A **stereo (dissecting) microscope** provides a genuinely three-dimensional view by giving each eye its own objective and eyepiece, i.e. two separate optical paths. It works on light reflected from the sample rather than transmitted through it, so it suits thick, solid, opaque specimens at low magnification (about 20x to 50x) — examining coins and artefacts, microsurgery, and viewing crystals. A **scanning probe microscope** abandons lenses altogether: a probe tip mounted on a cantilever is brought extremely close to the surface and tracks across it atom by atom. Forces between tip and surface deflect the cantilever, and a laser measures that deflection to reconstruct the image on a computer — used to study electrical and magnetic properties at the nanoscale.

2.5 Parts of the light microscope

For routine histology the compound light microscope is the workhorse. Its parts fall into optical, mechanical and illumination groups:

Group	Parts	Function
Optical	Eyepiece (ocular), objective lenses, nosepiece	Magnify the specimen; the nosepiece rotates objectives into place
Mechanical	Stage, stage clips, arm, base, head, focus system	Hold and position the slide; support the instrument; adjust focus
Illumination	Illuminator, condenser, diaphragm	Provide and concentrate light; control the amount reaching the specimen

ADVANTAGES & LIMITS OF LIGHT MICROSCOPY

Strengths: direct, real-colour imaging; fast; adaptable to almost any sample; easily fitted with a digital camera.

Limitations: relatively low resolution compared with electron microscopy, and the instrument is fragile and needs careful handling and maintenance.

3 Tissue Processing & Staining

A microscope is only as good as the specimen on the slide. Histological technique is the set of procedures that turns a fragment of living tissue into a thin, stained, permanent section. The guiding aim is faithfulness: the tissue on the slide should have the same structure and molecular composition it had in the body. The preparation itself can distort tissue or wash components away, so each step is designed to minimise that loss.

Techniques divide into **histological** methods, concerned with identifying cells and the structures the ECM builds, and **histochemical** methods, which localise specific cellular structures by exploiting a unique enzymatic activity present in them.

3.1 The seven steps of tissue processing

Preparing a paraffin section for light microscopy follows a fixed sequence. Each step prepares the tissue for the next.

#	Step	What happens and why
1	Fixation	Preserves structure and molecular composition and prevents autolysis (self-digestion). Done chemically (immersion) or by intravascular perfusion. Fixatives include formol-saline, formol-calcium, Zenker's, Bouin's, Carnoy's and Helly's fluids.
2	Dehydration	Water is drawn out by passing the tissue through graded ethanol of rising strength — typically 50%, 70%, 90% then 100%, about one hour in each.
3	Clearing	Ethanol is replaced by a solvent miscible with both alcohol and wax; the tissue becomes transparent. Xylene is usual (three changes); alternatives are toluene, histoclear and cedarwood oil.
4	Infiltration	The tissue is impregnated with molten embedding medium (paraffin wax or plastic resin) to give it a firm, uniform consistency.
5	Embedding	The correctly oriented specimen is enclosed in a support medium. Paraffin melted at 58–60°C fills every space; on cooling the tissue-plus-wax block hardens.
6	Sectioning	A microtome cuts the block with a steel or glass blade into thin sections 1–10 µm thick, which are floated on a warm water bath and lifted onto a glass slide.
7	Staining	Dyes are applied so that otherwise colourless structures become visible and contrast with one another.

3.2 Staining: acidic vs basic dyes

Most routine staining exploits a simple chemical attraction between charged dyes and charged tissue components. The naming is a classic exam trap, because each dye stains the *opposite* component to its own name.

Dye class	Stains which component?	Structure stained	Examples
Acidic dyes	The basic components of the cell	Cytoplasm	Eosin, orange G, acid fuchsin
Basic dyes	The acidic components of the cell	Nucleus (nucleic acids)	Haematoxylin, toluidine blue, alcian blue, methylene blue

HIGH-YIELD: WHY H&E LOOKS THE WAY IT DOES

In the standard **haematoxylin and eosin (H&E)** stain, basic haematoxylin binds the acidic nucleic acids of the nucleus and colours it blue-purple, while acidic eosin binds basic cytoplasmic proteins and colours the cytoplasm pink. Blue nucleus, pink cytoplasm — the appearance you will meet on almost every slide in this guide.

3.3 The H&E; staining protocol

The steps run 'downhill' to water, stain, then 'uphill' back to xylene for mounting:

- Dewax the section in xylene, then rehydrate through decreasing alcohol (high to low) to remove the xylene, and rinse in water.
- Stain in haematoxylin for about 5 minutes; rinse in running water.
- Differentiate in 1% acid alcohol to remove excess stain.
- 'Blue' the section by washing in running tap water for about 5 minutes.
- Counterstain in eosin for about 2 minutes; wash in three changes of water.
- Dehydrate through increasing alcohol (low to high), clear in two changes of xylene, air-dry, then mount and apply a coverslip.

4 Epithelial Tissue

Epithelial tissue is a sheet of cells attached to one another by specialised junctions, which covers every body surface and lines every cavity. Because it forms these boundaries, almost every substance the body takes in or gives off must cross an epithelium. Like all tissues it consists of cells plus a small amount of extracellular matrix, and it carries out protection, secretion, absorption, filtration, excretion and sensory reception.

4.1 General characteristics

- **Cellularity** — epithelium is almost entirely cells, with very little matrix between them.
- **Polarity** — the cells have a distinct free (apical) surface and an attached (basal) surface, and the two behave differently.
- **Attachment to a basement membrane** — the basal surface rests on a supporting layer that anchors it to the underlying connective tissue.
- **Avascular but innervated** — epithelium contains no blood vessels and is fed by diffusion from the connective tissue below, yet it is supplied with nerves.
- **Regeneration** — epithelial cells divide readily to replace those that are lost or damaged.
- **Support** — it is always underpinned by connective tissue.

4.2 The basement membrane

Epithelial cells do not sit directly on connective tissue. They rest on a **basal (reticular) lamina**, itself a connective-tissue product, which is bound in turn to the underlying areolar connective tissue. Taken together, these layers form the **basement membrane** — the interface that both anchors the epithelium and filters what passes between it and the tissue beneath.

4.3 Surface specialisations and cell junctions

The three surfaces of an epithelial cell carry different structural adaptations.

Surface	Specialisation	Function
Apical	Microvilli (brush border)	Greatly increase area for absorption or secretion
	Cilia	Motile projections that sweep fluids and mucus across the surface
	Stereocilia; flagella; glycocalyx	Long microvilli; whip-like motility; protective glycoprotein coat
Lateral	Tight (occluding) junctions – zonulae occludens	Seal the gap between cells so materials cannot leak between them
	Adhering junctions – zonulae adherens	Belt-like anchorage that binds neighbouring cells
	Gap (communicating) junctions	Channels that let ions and small molecules pass cell to cell
Basal	Basement membrane	Attaches the epithelium to underlying connective tissue

Surface	Specialisation	Function
	Desmosomes	Strong 'spot-weld' attachments that maintain sheet integrity

4.4 Classifying surface epithelia

Covering and lining epithelia are named by combining two features: the **shape** of the surface cells and the **number of layers**. Shape may be squamous (thin and flat), cuboidal (roughly square) or columnar (tall and rectangular). A single layer is **simple**; two or more layers make it **stratified**. A few types need special mention: pseudostratified epithelium looks layered because its nuclei sit at different heights, but every cell actually touches the basement membrane; transitional epithelium can stretch and change shape.

Type	Structure	Main function	Typical location
Simple squamous	Single layer of flat cells (like a fried egg)	Filtration, diffusion, osmosis	Lung air sacs, capillary walls, Bowman's capsule, lining of vessels
Simple cuboidal	Single layer of cube-shaped cells, central round nuclei	Secretion and absorption	Ovary surface, kidney tubules, ducts of pancreas and liver
Simple columnar	Single layer of tall cells; may bear cilia/microvilli; goblet cells secrete mucus	Absorption, secretion, protection	Lining of stomach, intestine and uterus
Pseudostratified columnar	Appears layered but all cells reach the base; often ciliated with goblet cells	Protection, secretion, moving mucus	Lining of the respiratory tract and some ducts
Stratified squamous	Many layers of flattened cells (keratinised or non-keratinised)	Protection against abrasion	Epidermis, mouth, throat, vagina, anal canal
Stratified cuboidal	Two or three layers of cuboidal cells	Protection	Ducts of sweat, salivary and mammary glands
Stratified columnar	Many layers of columnar cells	Protection and secretion	Vas deferens, part of male urethra, parts of the pharynx
Transitional	Many layers of varied shape that flatten on stretching	Distensibility and protection	Urinary bladder, ureters, part of the urethra

4.5 Glandular epithelium

Some epithelial cells specialise in producing a fluid secretion; collectively they form the glands. They synthesise, store and release products that are not used by the cell itself but serve the rest of the body. Glands develop in foetal life when a covering epithelium proliferates and grows down into the connective tissue beneath. Large multicellular glands (salivary glands, pancreas) contrast with unicellular glands such as the isolated goblet cell.

Exocrine vs endocrine

	Exocrine (ducted)	Endocrine (ductless)
Duct	Keeps its connection to the surface through a duct	Loses its surface connection during development

	Exocrine (ducted)	Endocrine (ductless)
Secretion route	Product travels along the duct to a surface (skin, gut)	Hormones released directly into the bloodstream
Parts	Secretory portion + duct system	Each cell sits in direct contact with a blood capillary
Examples	Sweat, salivary and mammary glands, pancreas (exocrine part)	Thyroid, parathyroid, adrenal, pituitary (adenohypophysis)

How exocrine glands secrete (functional classification)

Mechanism	What is released	Example
Merocrine / eccrine	Product leaves by exocytosis with no loss of cell substance	Pancreas, salivary glands
Apocrine	The apical cytoplasm is shed along with the product	Female mammary gland
Holocrine	The entire cell breaks down and is discharged with its product	Sebaceous glands of the skin

Exocrine glands are also classified structurally: by their duct system as **simple** (unbranched) or **compound** (branched), and by their secretory portion as **tubular**, **acinar/alveolar** (flask-shaped) or **tubuloacinar**. Their secretion may be **serous** (watery, enzyme-rich) or **mucous** (viscous, protective and lubricating). Endocrine glands, meanwhile, are arranged as trabecular cords (adenohypophysis, parathyroid, adrenal), follicular spheres (thyroid) or disseminated cells scattered in another organ (Leydig cells of the testis, the islets of Langerhans in the pancreas).

4.6 Special epithelia

- **Myoepithelium** — muscle-like, contractile branched cells containing actin, lying between glandular cells and the basement membrane; they squeeze secretion out of the gland.
- **Neuroepithelium** — sensory cells, both non-neuronal (taste cells, hair cells) and neuronal (olfactory sensory cells, the rods and cones of the retina).
- **Seminiferous epithelium** — the germinal epithelium of the testis, combining spermatogenic (germ) cells with supporting Sertoli cells.

4.7 Renewal and clinical correlations

Epithelium is constantly renewed — new cells arise from existing epithelial cells, from connective-tissue stem cells and from bone-marrow-derived stem cells. This same capacity for rapid division makes epithelium a common site of disease.

CLINICAL CORRELATIONS

Carcinomas. More than 90% of all tumours are of epithelial origin, including squamous-cell and basal-cell carcinomas; malignant tumours of glandular epithelium are called **adenocarcinomas**. Uncontrolled epithelial growth (neoplasia) is not always cancer — it can be reversible.

Metaplasia. One mature epithelium may reversibly convert into another. In heavy smokers the ciliated pseudostratified lining of the bronchi becomes stratified squamous; chronic vitamin-A deficiency drives a similar change in the bronchi and bladder. Metaplasia can also occur in connective tissue.

Others. Hyperkeratosis (a genetic disorder of impaired desquamation with scaly skin) and epidermolysis bullosa illustrate what happens when epithelial adhesion and renewal fail.

5 Connective Tissue

Connective tissue fills the spaces between the other tissues and, in doing so, helps build the organs. On a slide it is the tissue you see *between* other layers — for example between the epithelium and the muscle in the wall of a hollow organ. Its job is to provide general structure, mechanical strength, space-filling bulk that sculpts the body's shape, and both physical and metabolic support for more specialised tissues.

Three structural demands shape connective tissue, each met by a particular material: **tensile strength** to resist pulling and tearing, supplied by strong collagen fibres; **elasticity** to spring back after distortion, supplied by rubber-like elastin fibres; and **volume or bulk**, supplied by water-binding glycoproteins and complex carbohydrates that form a wet gel.

5.1 Origin and composition

Connective tissue arises from **mesenchyme**, an embryonic tissue of elongated, undifferentiated mesenchymal cells. It is built from two ingredients: the **extracellular matrix** (ground substance plus protein fibres) and the **connective-tissue cells** themselves. Its broad functions are to support organs and cells, to act as a medium for exchanging nutrients and wastes between blood and tissues, to protect against microorganisms, to repair damage and to store fat.

5.2 The extracellular matrix

The ECM is the major constituent of connective tissue and a mixture of protein fibres embedded in **ground substance** — a colourless, highly hydrophilic, transparent gel of glycosaminoglycans, proteoglycans and glycoproteins. The ground substance lubricates, resists compression and helps bar the invasion of tissues by foreign agents. Three fibre types run through it:

Fibre	Made of	Key features
Collagen	Type I and III collagen (of 25+ types)	Great tensile strength; the most abundant protein in the body (about 30% of dry weight); found in bone, skin, cartilage and tendon
Reticular	Mainly type III collagen	Very thin (0.5–2.0 μm); high carbohydrate content; stain black with silver salts; form the framework of organs and glands
Elastic	Elastin core with fibrillin microfibrils	Coiled, branching, 0.2–1.0 μm ; stretch up to 150% of resting length; need special staining to be seen

HIGH-YIELD

Collagen is the most abundant structural protein of the ECM. It is made in a two-stage process spanning both the inside and the outside of the cell, and it gives tissues both flexibility and strength.

5.3 The cells of connective tissue

Connective-tissue cells fall into two groups by residence. **Fixed cells** originate locally and stay put; **transient (wandering) cells** arise elsewhere — chiefly from haematopoietic stem cells in the bone marrow — and pass through only temporarily, typically as part of defence and inflammation.

Fixed cells

Cell	Origin & features	Role
Fibroblast	From mesenchyme; the predominant cell; oval nucleus, 2+ nucleoli; active cells are spindle-shaped with abundant rough ER and Golgi	Secrete, maintain and recycle the matrix (procollagen and ECM); divide mainly during wound healing
Pericyte	From mesenchyme; lie along capillaries; contain actin, myosin and tropomyosin	Contractile; can differentiate into endothelial and smooth-muscle cells during vessel growth and repair
Adipocyte (fat cell)	From mesenchyme/fibroblasts; fully differentiated, surrounded by a basal lamina	Synthesise, store and release fat
Fixed macrophage	Resident phagocyte of the connective tissue	Local clearance of debris and defence

Adipocytes come in two forms: **unilocular** cells (white adipose tissue) hold a single large fat droplet and carry receptors for insulin, growth hormone, noradrenaline and glucocorticoids; **multilocular** cells (brown adipose tissue) are smaller and store fat as many small droplets around a central nucleus.

Transient cells

Cell	Role
Macrophage	Principal phagocyte; removes large particles, presents antigen to lymphocytes and secretes cytokines. Arises from blood monocytes. Known by special names in some organs — Kupffer cells (liver), osteoclasts (bone), microglia (CNS)
Mast cell	One of the largest CT cells; cytoplasm packed with metachromatic granules of heparin and histamine; sits near small vessels and drives allergic/inflammatory responses
Plasma cell	Antibody factory derived from activated B lymphocytes; eccentric 'clock-face' nucleus and deeply basophilic cytoplasm; abundant at wound sites and chronic inflammation
Lymphocytes	T cells (cell-mediated immunity), B cells (become plasma cells for humoral immunity) and NK cells (kill tumour and infected cells)
Granulocytes	Neutrophils (phagocytose and kill bacteria; dead neutrophils form pus); eosinophils (attack parasites and moderate allergy); basophils (blood-borne relatives of mast cells)

REMEMBER: THE FOUR MAJOR CT CELLS

The four cells most emphasised are the **fibroblast** (arises locally), the **macrophage**, the **mast cell** and the **plasma cell**. The last three ultimately originate from haematopoietic stem cells in the bone marrow, circulate in the blood, then settle in connective tissue to do their work.

5.4 Classification of connective tissue

Connective tissue is classified by the proportion of cells to fibres and by the arrangement and type of those fibres. At the top level it splits into embryonic tissue, connective tissue proper (the adult form) and specialised connective tissue.

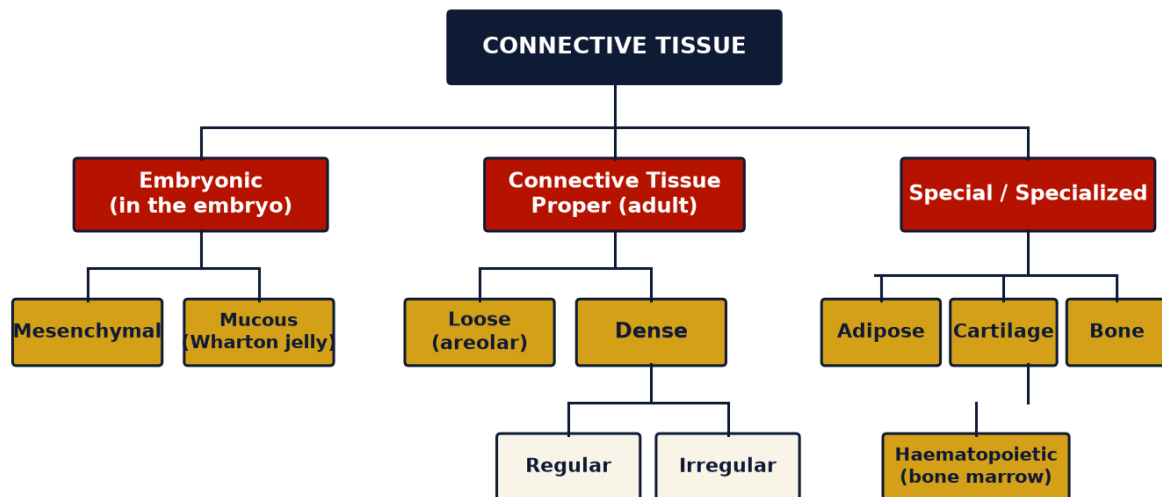


Figure 5.1 — Classification of connective tissue (original schematic).

Embryonic connective tissue

A loose tissue of early development, in two forms. **Mucous tissue (Wharton's jelly)** is the jelly-like main constituent of the umbilical cord, holding a few collagen fibres and large stellate fibroblasts. **Mesenchymal tissue**, found only in the embryo, is a gel of scattered reticular fibres and pluripotent star-shaped cells in active mitosis.

Connective tissue proper

The adult form is remarkably diverse because the ratio of cells to fibres varies from place to place. It divides into loose and dense types:

Type	Cells vs fibres	Properties & location
Loose (areolar)	More cells than collagen fibres	Well vascularised, flexible, not very stress-resistant; the most abundant CT; fills spaces beneath the skin and around vessels, nerves and muscles
Dense regular	More collagen than cells; fibres parallel	Resists pull in one direction; tendons, ligaments, cornea
Dense irregular	More collagen than cells; fibres random	Resists pull from many directions; dermis of the skin
Reticular	Network of reticular fibres	Delicate meshwork forming the stroma of lymph nodes, spleen, bone marrow and liver
Elastic	Irregular elastic fibres	Provides recoil; ligaments of the vertebral column, walls of large arteries

Specialised connective tissue

Four tissues have properties beyond those of the ordinary adult forms: **adipose tissue**, **cartilage**, **bone** and **haematopoietic tissue (bone marrow)**. Bone and cartilage are covered in detail in the next section.

5.5 Adipose tissue in focus

Adipose tissue is specialised connective tissue dominated by adipocytes, found throughout the body either isolated or in groups within loose connective tissue. It represents roughly 15–20% of body weight in men of normal weight and 20–25% in women. It is the body's largest energy store, holding fuel as triglyceride — unlike the liver and skeletal muscle, which store energy as glycogen. Adipocytes are now recognised as key regulators of energy metabolism and as an endocrine and signalling organ, releasing hormones and other factors.

CLINICAL IMPORTANCE

With the worldwide rise in obesity and its links to diabetes and heart disease, the adipocyte has become the most widely studied cell of connective tissue. White adipose tissue is evenly spread in the newborn; as a child matures it recedes from some regions and builds up in others, its distribution partly controlled by sex hormones (for example in the breasts and thighs).

6 Bone & Cartilage

Bone (osseous tissue) is a hard, specialised connective tissue with an internal honeycomb-like matrix that gives it rigidity. Like every connective tissue it is cells embedded in a matrix — but here the matrix is mineralised, combining an organic framework of collagen (called ossein) with an inorganic component of calcium salts. That pairing lets bone be simultaneously hard, strong and surprisingly light.

6.1 Bone structure and composition

Bone is not a solid block. It is a flexible organic matrix (about 30%) impregnated with bound mineral (about 70%), woven together and continuously remodelled. The **organic matrix** is 90–95% elastic collagen (ossein) plus ground substance; the collagen supplies tensile strength and fracture resistance. The **inorganic mineral** is chiefly hydroxyapatite, calcium phosphate of nominal formula $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$, which supplies compressive strength and rigidity. The two effects are synergistic: collagen resists stretching, mineral resists crushing.

6.2 Compact and cancellous bone

Within any bone the tissue is woven into two patterns. The dense outer shell is **cortical (compact) bone**; the porous internal network is **cancellous (spongy or trabecular) bone**.

Feature	Cortical (compact)	Cancellous (spongy)
Location	Hard outer cortex of bones	Interior, at the ends of long bones, near joints, in vertebrae
Density	Dense; smooth, white, solid	Porous open-cell network; less dense
Share of bone mass	About 80% of adult skeletal mass	About 20% of mass, but ~10x the surface area of compact bone
Functional unit	The osteon (Haversian system)	The trabecula
Notes	Columns of cells around a central (osteonic) canal; Volkmann's canals link osteons at right angles	Highly vascular; houses red marrow where haematopoiesis occurs

Compact bone is covered by a **periosteum** on its outer surface and an **endosteum** on its inner surface; the endosteum marks the boundary between compact and cancellous bone. In spongy bone, thin osteoblast formations covered by endosteum build an irregular network of spaces (trabeculae) whose alignment follows the mechanical loads the bone must bear. These spaces contain marrow and haematopoietic stem cells that give rise to red cells, white cells and platelets.

6.3 The three bone cells

Cell	Origin	Function
Osteoblast	Mononucleate, on bone surfaces	Build bone: lay down collagen and secrete osteoid (unmineralised type-I-collagen matrix), then deposit calcium phosphate; also make hormones such as prostaglandins

Cell	Origin	Function
Osteocyte	An osteoblast trapped in the matrix it made	Maintain bone; sit in spaces called lacunae, extending processes through canaliculi to contact other cells via gap junctions
Osteoclast	Large, multinucleate; from a monocyte lineage	Resorb bone: secrete enzymes (e.g. tartrate-resistant acid phosphatase) into resorption pits (Howship's lacunae); central to calcium homeostasis

HIGH-YIELD: REMODELLING

Bone is built and rebuilt throughout life. Osteoblasts lay down new osteoid that mineralises into bone; an osteoblast that becomes walled in turns into an osteocyte, while those left on the surface become protective bone-lining cells. Osteoclasts counterbalance this by resorbing bone. The continuous push-pull of osteoblast building and osteoclast resorption is **bone remodelling**, and it also fine-tunes blood calcium.

6.4 Cartilage

Cartilage is a resilient, smooth connective tissue — semi-transparent, non-porous and usually wrapped in a tough fibrous **perichondrium**. It is far stiffer than muscle but neither as hard nor as rigid as bone, and it typically grows faster than bone. It caps the ends of long bones as articular cartilage and forms parts of the rib cage, neck, bronchial tubes and intervertebral discs; because of its rigidity it also holds tubes such as the trachea open. Its matrix is glycosaminoglycans, proteoglycans, collagen and sometimes elastin, produced by cells called **chondrocytes**.

Crucially, cartilage has no blood vessels and no nerves, so it is insensitive and its chondrocytes are fed only by diffusion — movement of the tissue helps drive fluid flow that carries nutrients in. This is why cartilage turns over slowly and repairs only poorly. Three types are recognised by the balance of collagen and proteoglycan:

Type	Cells & matrix	Location
Hyaline	Fewer cells, more intercellular space	Nose, ends of long bones, trachea, parts of larynx, smaller airways
Elastic	Cells closer together, less space; rich in elastin	External ear (pinna), parts of the larynx
Fibrocartilage	Fewest cells, most intercellular space; tough	Intervertebral discs of the spine, the menisci of the knee

7 Muscle Tissue

Muscle tissue is made of contractile cells called myocytes. It develops mainly from mesoderm (with a few exceptions — the intraocular smooth muscles come from ectoderm), its embryonic precursors being myoblasts derived from mesenchyme. Its signature is the ability to shorten forcefully, powered by the interaction of two protein filaments, thin **actin** and thick **myosin**.

7.1 Shared properties

- **Contractility** — the universal muscle property, requiring the myofilaments actin and myosin.
- **Excitability** — responds to electrical, hormonal and mechanical stimuli.
- **Extensibility** — can be stretched, and **elasticity** — recoils to its resting shape when tension is released.

Through these properties muscle produces movement, provides stability (supporting the skeleton, stabilising joints, maintaining posture) and controls body passages — setting vessel diameter by vasoconstriction and vasodilation, moving food by peristalsis, and closing openings with sphincters.

7.2 The three types of muscle

Feature	Skeletal	Cardiac	Smooth
Striations	Striated	Striated	Non-striated
Cell shape	Long, cylindrical, multinucleate	Branched, elongated, central nucleus	Fusiform (tapered), one central nucleus
Nuclei	Many, at the periphery	Usually one, central	One, central
Control	Voluntary (mostly)	Involuntary; self-exciting	Involuntary
Location	Attached to skeleton, moves the eyes	Wall of the heart	Walls of vessels and hollow organs

The presence or absence of striations reflects how the filaments are arranged. In **striated** muscle (skeletal and cardiac) actin and myosin are packed into regular repeating units called **sarcomeres**. In **non-striated** smooth muscle the same proteins are present but arranged irregularly, without sarcomeres, so no banding shows.

7.3 Skeletal muscle fine structure

A skeletal muscle fibre is a large multinucleated cell (10–100 µm across) with its nuclei pushed to the periphery. Its membrane, the **sarcolemma**, dips inward as **transverse (T) tubules** that carry electrical signals deep into the fibre. Its cytoplasm, the **sarcoplasm**, is packed with rod-like **myofibrils** and is rich in myoglobin (which stores oxygen) and glycogen (fuel). Wrapped around each myofibril is the **sarcoplasmic reticulum**, a specialised smooth ER whose terminal cisternae line the T tubules and release the calcium that triggers contraction.

The contractile proteins

Protein	Type	Role
Actin	Contractile (thin)	Two coiling strands; slides against myosin

Protein	Type	Role
Myosin	Contractile (thick)	Shaft with globular heads that pull on actin
Tropomyosin	Regulatory	Blocks actin's binding sites when relaxed
Troponin	Regulatory	Calcium-binding switch that controls contraction
Titin	Elastic	Runs through myosin and anchors it to the Z line

7.4 The sarcomere and its bands

Myofilaments are organised into sarcomeres — the functional units of striated muscle — which interlock end to end to form the myofibril. The overlap of thick and thin filaments creates the pattern of bands that gives striated muscle its name:

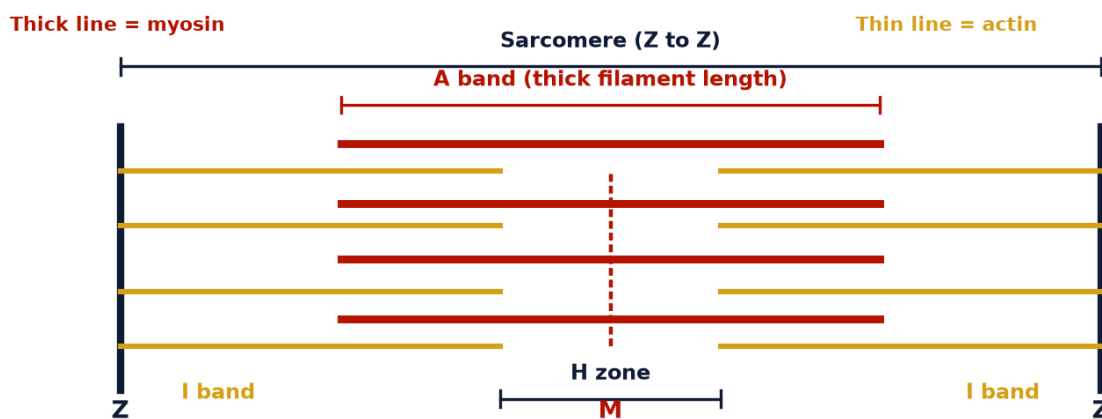


Figure 7.1 — The sarcomere and its bands (original schematic).

Band / line	What it is
Z band (Z line/disc)	Anchors the thin filaments and bounds the sarcomere; a sarcomere is the region between two Z bands
A band	Dark band = the full length of the thick myosin filaments (including where actin overlaps). Memory aid: dArk has an A
I band	Light band of thin actin only, straddling the Z line and lying between A bands. Memory aid: lIght has an I
H zone	Lighter central region of the A band containing myosin only (no actin overlap)
M band	Fine dark line at the centre of the H zone where myosin-binding proteins attach

7.5 Skeletal muscle fibre types

Type	Nickname	Properties	Example
Type I	Slow-twitch, slow oxidative	Fatigue-resistant; small red fibres	Postural back muscles

Type	Nickname	Properties	Example
Type IIA	Fast oxidative-glycolytic	Fatigue-resistant; intermediate size; high sustained power	Sustained powerful movement
Type IIB	Fast glycolytic	Store much glycogen; fatigue quickly (lactic acid); large pink fibres	Brief, powerful bursts

7.6 Smooth muscle

Smooth muscle is non-striated and involuntary, controlling internal organs and vessels. It lines the vasculature, the GI tract, the respiratory and urinary tracts, the female reproductive tract, the iris of the eye and the piloerector muscles of hair follicles. Its cells are fusiform with a single central nucleus, arranged in parallel. Actin and myosin are present but not in ordered sarcomeres; instead the filaments attach to **dense bodies** scattered in the cytoplasm and on the membrane, which act like the Z discs of striated muscle, with intermediate filaments linking dense bodies together. Smooth muscle has little sarcoplasmic reticulum and no T tubules, an arrangement that favours slow, sustained contraction. Impulses spread cell to cell through gap junctions, enabling coordinated waves such as peristalsis.

Type	Coupling	Behaviour & location
Single-unit	Cells electrically coupled by gap junctions (a functional syncytium)	Slow, wave-like contraction; the common type — blood vessels and most visceral organs, often in circular and longitudinal layers
Multi-unit	Cells separated by basement membrane; no gap junctions	Each cell contracts independently; largest arteries and airways, piloerector muscles, iris of the eye

7.7 Cardiac muscle

Cardiac muscle is striated but, unlike skeletal muscle, works autonomously through its own pacemaker cells. It is found almost exclusively in the heart (with a few cells in the aorta and superior vena cava). Its cells have a central nucleus and branch so that one cell connects to many others through **intercalated discs**; they are arranged in spiralling woven layers that wring out the heart chambers as they contract, and the ventricular muscle is much thicker than the atrial. The cells are dense with mitochondria to meet their relentless energy demand, and their actin/myosin arrangement mirrors skeletal muscle, forming the same sarcomeric striations.

HIGH-YIELD: THE INTERCALATED DISC

Unique to cardiac muscle, intercalated discs join cardiomyocytes end to end. They combine **desmosomes**, which anchor one cell mechanically to the next, with **gap junctions**, which act as electrical synapses so impulses spread rapidly and the whole muscle contracts in synchrony. Cardiac muscle can contract without any nerve supply thanks to pacemaker cells in the sinoatrial (SA) node, while the autonomic nervous system merely adjusts heart rate and force.

8 Nervous Tissue

Nervous tissue is built from two principal kinds of cell. The **neuron** is the structural and functional unit — an electrically excitable cell that receives, processes and transmits electrical signals to and from other parts of the nervous system through its processes. The **supporting (neuroglial) cells** lie close to the neurons but do not themselves conduct electrical signals; they nourish, insulate and protect. Neurons are classified by structure and function as sensory neurons, motor neurons and interneurons.

8.1 The functional parts of a neuron

Part	Function
Dendrites	Receive incoming signals and carry them toward the cell body
Cell body (soma)	Houses the nucleus and drives the cell's metabolic activities
Axon	Conducts impulses away from the cell body toward target cells
Synaptic junctions	Specialised contacts that pass impulses between neurons, or from a neuron to an effector such as muscle or gland

8.2 Neurons of the cerebral cortex

The dominant cells seen on a section of cerebral cortex are the **pyramidal cells**. These act as upper motor neurons: they synapse onto the lower motor neurons that contact muscle directly to initiate contraction. The axons of pyramidal cells descend through the white matter of the cerebrum into the three parts of the brainstem — the midbrain, the pons and the medulla. Within the medulla most axons cross to the opposite side of the body in a process called **decussation**, then continue down the white matter of the spinal cord before synapsing with interneurons and lower motor neurons in the grey matter.

8.3 The cerebellar cortex

The cortex of the cerebellum is organised into three layers, and recognising them is a classic histology task:

Layer	Position	Contents
Molecular layer	Outer	Relatively few cells, with granules interspersed throughout
Purkinje cell layer	Central	A single row of large, easily recognised Purkinje cells with extensive dendritic trees
Granular layer	Inner	Densely packed granule cells

8.4 Nerves in section

A **nerve** is a bundle of axons running in parallel in the peripheral nervous system (PNS); it therefore contains axons but no neuronal cell bodies. In the PNS the insulating **myelin** around axons is produced by **Schwann cells**. A nerve looks completely different depending on the plane of section:

Plane	Appearance
Longitudinal section	Axons run as darker-staining purple lines; the paler layers flanking each axon are myelin, often many times thicker than the axon itself
Transverse section	Each myelinated axon appears as a little circle; the thin central axon is a purple dot inside a paler ring of myelin (stained dark with osmium). Axons are grouped into bundles called fascicles, each wrapped by an endoneurium of loose reticular connective tissue

EXAM POINTER

If a slide of 'nerve' shows no cell bodies, that is expected — the bodies sit in the CNS grey matter or in peripheral ganglia, not in the nerve trunk. Distinguish the two planes at a glance: parallel purple lines = longitudinal; rings-within-rings = transverse.



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ABOUT THIS GUIDE

This guide reorganises and rewrites introductory histology material into a single continuous reference covering microscopy and tissue technique, the four basic tissues, and bone and cartilage. Diagrams have been redrawn from scratch. It is intended purely as a revision aid to sit alongside your own notes and your lecturer's current outline.

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